

1. AIM

To design and study a lithography mask pattern for shallow trench isolation, illustrating line space structures within a defined boundary for device isolation for given masks.

2. TOOL

Semulator3D

3. THEORY

Shallow Trench Isolation (STI) is a semiconductor fabrication technique used to electrically isolate adjacent devices. In this process, shallow trenches are formed in the silicon substrate and filled with an insulating material. Lithography is used to define the regions where the trenches are to be formed.

In this assignment, two lithography mask designs are studied. Design 1 consists of a $500\text{ nm} \times 400\text{ nm}$ boundary containing four vertical mask openings, each 40 nm wide and separated by 60 nm spacing. The openings form a regular line-space grating oriented along the Y-axis. Design 2 consists of a $500\text{ nm} \times 500\text{ nm}$ boundary with a 60 nm wide horizontal bar and a 60 nm wide vertical bar, both extending across the complete boundary and intersecting at the center.

These patterns illustrate how lithographic dimensions such as opening width, spacing, orientation, and alignment are used to define regions for device isolation. Precise control of these parameters is important for transferring the required pattern accurately during semiconductor fabrication.

4. DESIGN PARAMETERS

Table 1: assignment 1

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	Silicon/Silicon...		200			WAFER_1				
2		Conformal...	pad oxide deposition	/Oxides/Silicon...		4			WAFER_1	Both	1.0		
3		Conformal...	pad nitride deposition	/Nitrides/SiN		40			WAFER_1	Top	1.0		
4		Sequence	lithography						WAFER_1				
4.1		Planarizing...	spin on resist	/Organics/photo...		120			WAFER_1	Top			
4.2		Expose resist	Expose resist	/Organics/photo...		0	tx	Light	WAFER_1	Top			
5		Sequence	etch fins						WAFER_1				
5.1		Enhanced B...	pad nitride etch	/Nitrides/Org...		40			WAFER_1	Top	0.05		
5.2		Enhanced B...	pad oxide etch	/Nitrides/Org...		4			WAFER_1	Top	0.05		
5.3		Enhanced B...	silicon etch	/Nitrides/Org...		100			WAFER_1	Top	0.05		
6		Remove M...	resist strip	/Organics/					WAFER_1				
7		Conformal...	sti fill deposition	/Oxides/SiO2_H...		200			WAFER_1	Top	0.67		
8		Polish	sti oxide cmp						WAFER_1	Top			

Table 2: assignment 2

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	Silicon/Silicon...		200			WAFER_1				
2		Conformal...	pad oxide deposition	/Oxides/Silicon...		4			WAFER_1	Both	1.0		
3		Conformal...	pad nitride deposition	/Nitrides/SiN		40			WAFER_1	Top	1.0		
4		Sequence	lithography						WAFER_1				
4.1		Planarizing...	spin on resist	/Organics/photo...		120			WAFER_1	Top			
4.2		Expose resist	Expose resist	/Organics/photo...		0	cross_line	Light	WAFER_1	Top			
5		Sequence	etch fins						WAFER_1				
5.1		Enhanced B...	pad nitride etch	/Nitrides/Org...		40			WAFER_1	Top	0.05		
5.2		Enhanced B...	pad oxide etch	/Nitrides/Org...		4			WAFER_1	Top	0.05		
5.3		Enhanced B...	silicon etch	/Nitrides/Org...		100			WAFER_1	Top	0.05		
6		Remove M...	resist strip	/Organics/					WAFER_1				
7		Conformal...	sti fill deposition	/Oxides/SiO2_H...		200			WAFER_1	Top	0.67		
8		Polish	sti oxide cmp						WAFER_1	Top			

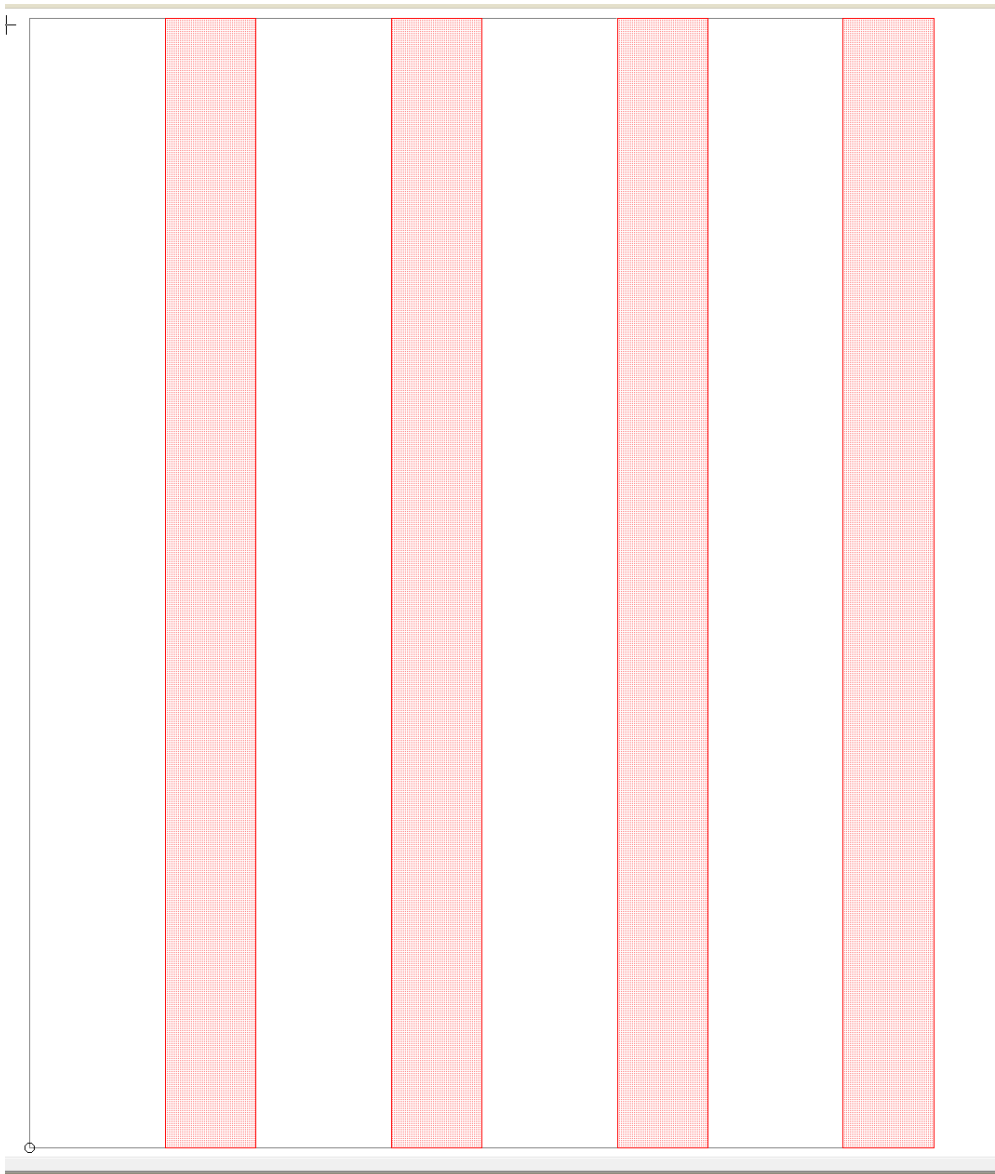


Figure 1: Assignment1

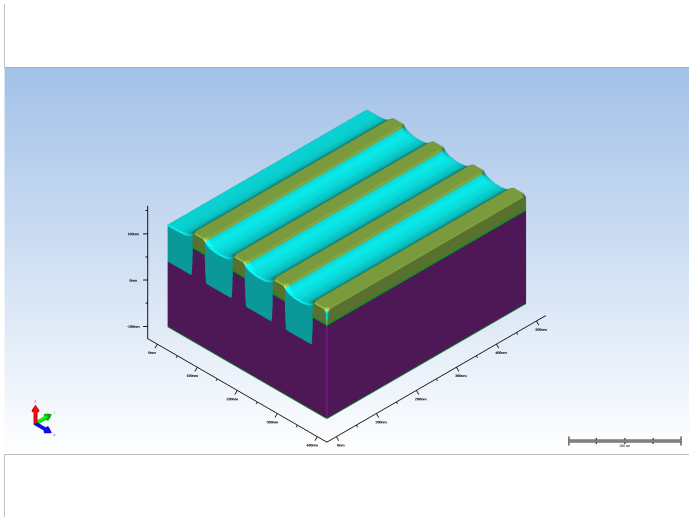


Figure 2: assignement1 Slide 11

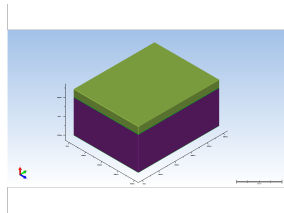
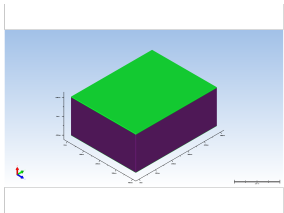
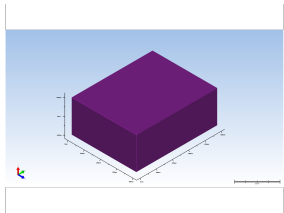


Figure 3: assignement1 Slide 1 Figure 4: assignement1 Slide 2 Figure 5: assignement1 Slide 3

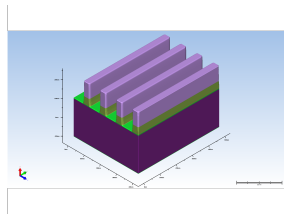
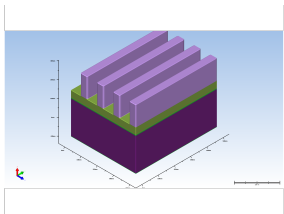
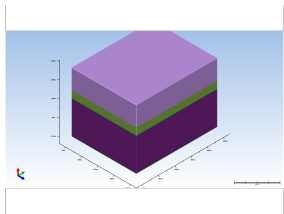


Figure 6: assignement1 Slide 4 Figure 7: assignement1 Slide 5 Figure 8: assignement1 Slide 6

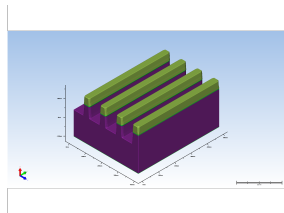
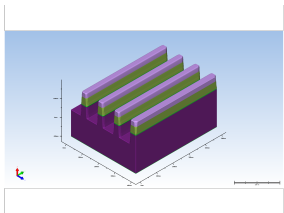
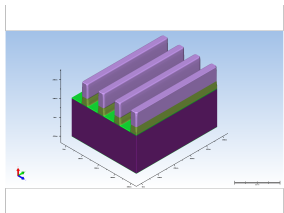


Figure 9: assignement1 Slide 7 Figure 10: assignement1 Slide 8 Figure 11: assignement1 Slide 9

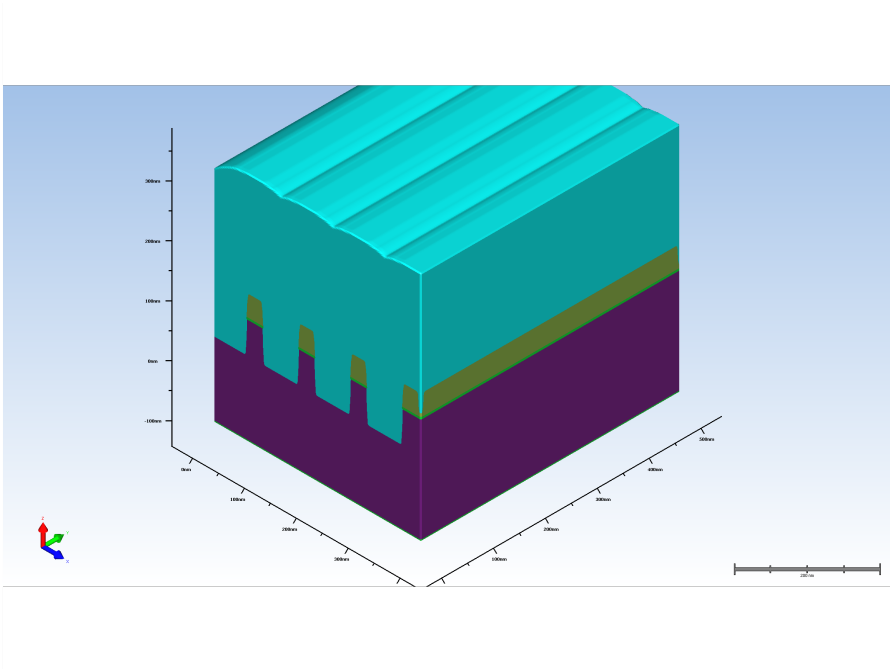


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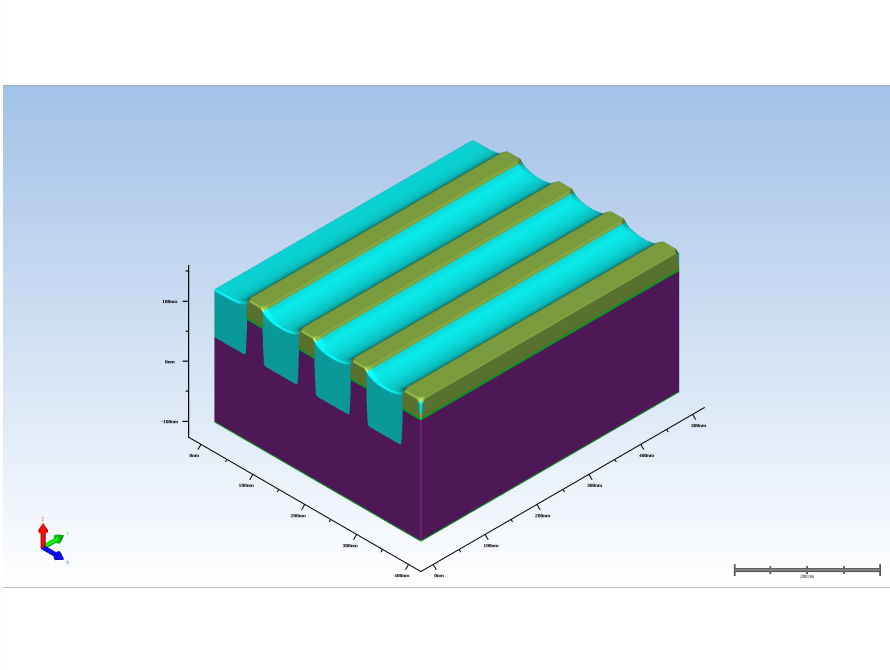


Figure 13: assignement1 Slide 11

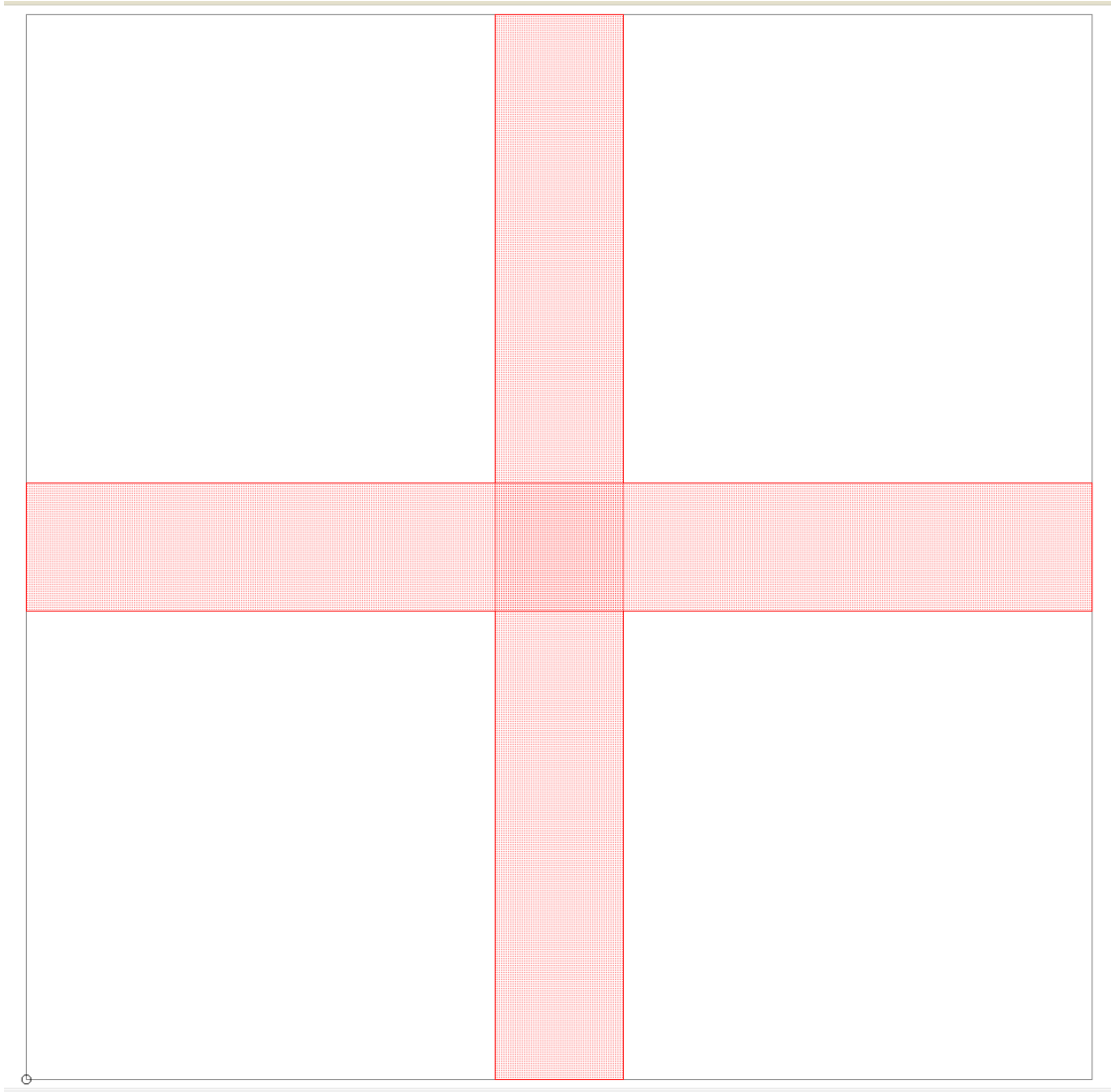


Figure 14: Assignment2

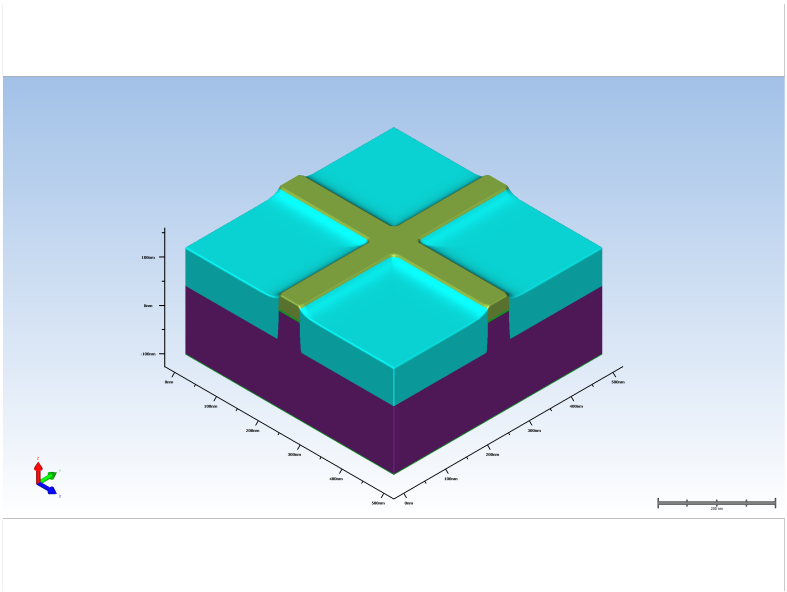


Figure 15: assignement2 Slide 11

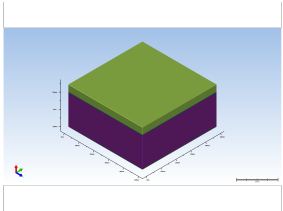
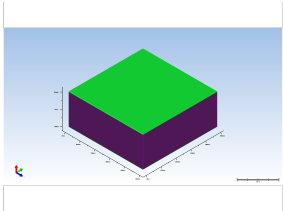
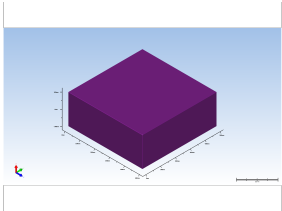


Figure 16: assignement2 Slide 1 Figure 17: assignement2 Slide 2 Figure 18: assignement2 Slide 3

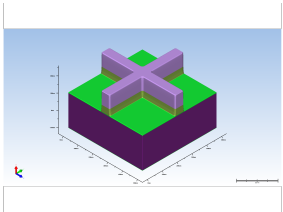
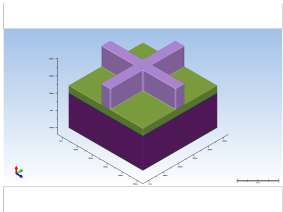
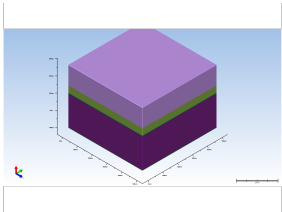


Figure 19: assignement2 Slide 4 Figure 20: assignement2 Slide 5 Figure 21: assignement2 Slide 6

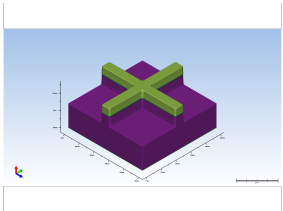
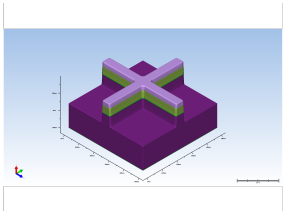
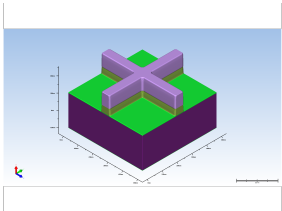


Figure 22: assignement2 Slide 7 Figure 23: assignement2 Slide 8 Figure 24: assignement2 Slide 9

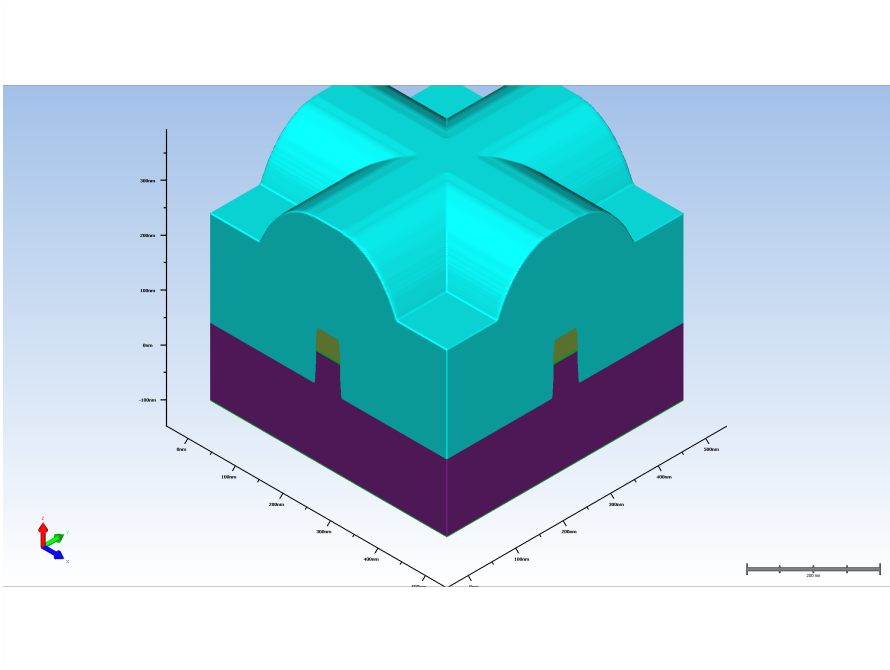


Figure 25: assignment2 Slide 10

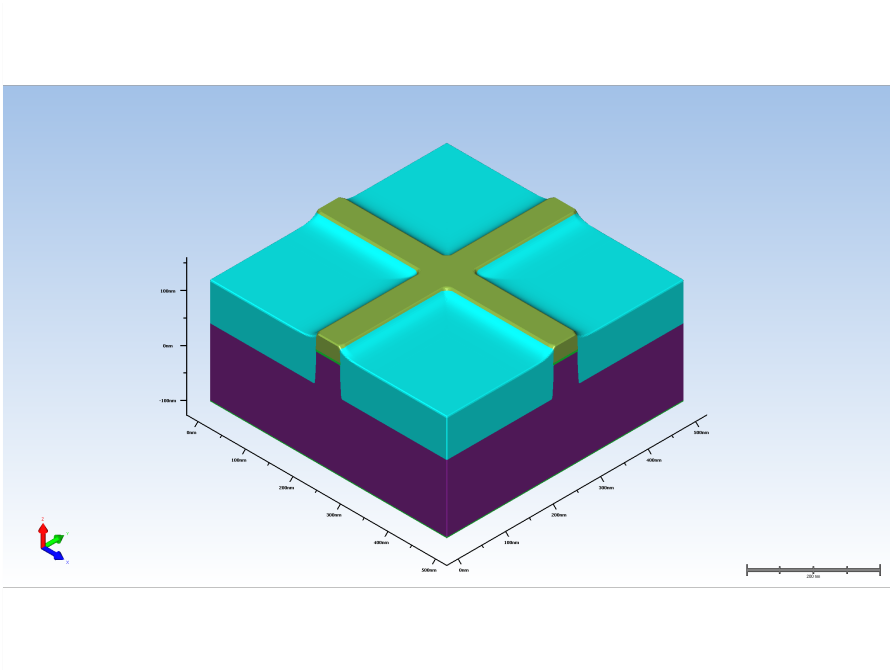


Figure 26: assignment2 Slide 11

6. OBSERVATION

Design 1 showed four uniform vertical line openings within the $500\text{ nm} \times 400\text{ nm}$ boundary. Each line was 40 nm wide with 60 nm spacing, producing a regular line-space grating along the Y-axis.

Design 2 showed a cross-shaped pattern within the $500\text{ nm} \times 500\text{ nm}$ boundary. The horizontal and vertical openings were each 60 nm wide, extended across the full boundary, and intersected at the center.

7. CONCLUSION

The lithography mask patterns for shallow trench isolation were successfully designed and studied using Simulator3D. The parallel line array and orthogonal cross pattern were created with the specified dimensions, spacing, orientation, and boundary limits. The experiment demonstrated how accurately defined mask openings are used to represent isolation regions during semiconductor fabrication.